Academic Programs

B.Tech. ? Biotechnology

Programme Educational Objectives (PEOs)

PEO-1: Graduates will have a strong foundation in fundamentals of mathematics, physical and biological sciences, Engineering sciences, to solve real time problems in health care, environment, Bioprocess Engineering and food security and successfully pursue higher studies

PEO-2: Graduates will have successful professional career by demonstrating good scientific and engineering breadth to comprehend the problems, conduct experiments, analyze the results and design novel products and solutions to the real life problems, promote entrepreneurship and skills in project and finance management related to Biotechnology and related domains

PEO-3: Graduates will be trained in biosafety, regulatory and Intellectual Property related issues in broader social context and sustainable development. Professional ethics, communication skills, team work skills, leadership and multidisciplinary approach to exhibit the ability to work effectively in multidisciplinary teams.

PEO-4: Graduates will be motivated to achieve academic excellence and pursue research to develop life ? long learning in a world of constantly evolving technologies associated with biotechnology and related domains to become Biotechnologist and Bioengineers to excel in academia, industry, or entrepreneurship in the biotechnology sector

Program Outcomes (POs)

Graduates will be able to:

Engineering knowledge: Apply knowledge of mathematics, basic sciences, computing, engineering fundamentals to develop solutions for biotechnology related problems. (WK1-WK4)

Problem analysis: Review research literature and demonstrate an ability to Identify, formulate and analyze complex problems in key areas of Biotechnology ? Bioprocess engineering, plant, medical and environmental biotechnology etc and related domains reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

Design/development of solutions: Design creative solutions for complex biotechnology and bioengineering problems and design / develop systems / components / processes to

meet identified
needs, conduct experiment, analyze and interpret data to get sustained
conclusion in biotechnology and
related domains for human welfare. (WK5).

Conduct investigations of complex problems:

Conduct investigations of complex biotechnology
and bioengineering problems using research-based knowledge including design of
experiments,
modelling, analysis & interpretation of data to provide valid conclusions.
(WK8).

Engineering tool usage: Demonstrate the skills to select and

apply appropriate modern
engineering and IT tools, software?s and equipment to experiment and analyze the
problems in
biotechnology, bioengineering and related domains. (WK2 and WK6).

The Engineer and The World: Analyze and evaluate societal and

environmental aspects while
solving complex engineering problems for its impact on sustainability with
reference to economy,
health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

Ethics: Apply ethical principles and commit to professional

ethics, understand human values,
diversity and inclusive behavior assessment, and adhere to national &
international laws. (WK9).

Individual and Collaborative Team work: Function effectively as

an individual, and as a member or
leader in diverse/multi-disciplinary teams.

Communication: Communicate effectively and inclusively within

the Professional community and
society at large, and being able to comprehend and write effective reports and
design documentation,
make effective presentations considering cultural, language, and learning
differences.

Project management and finance: Apply knowledge and understand

management principles and
economic decision-making in respective domain and apply these to one?s own work,
as a member and
leader in a team, and to manage projects in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the

preparation and ability to be

i) independent and life-long learner
ii) adaptable to new and emerging technologies and
iii) critical thinking in the broadest context of technological change in

respective and related domains. (WK8).